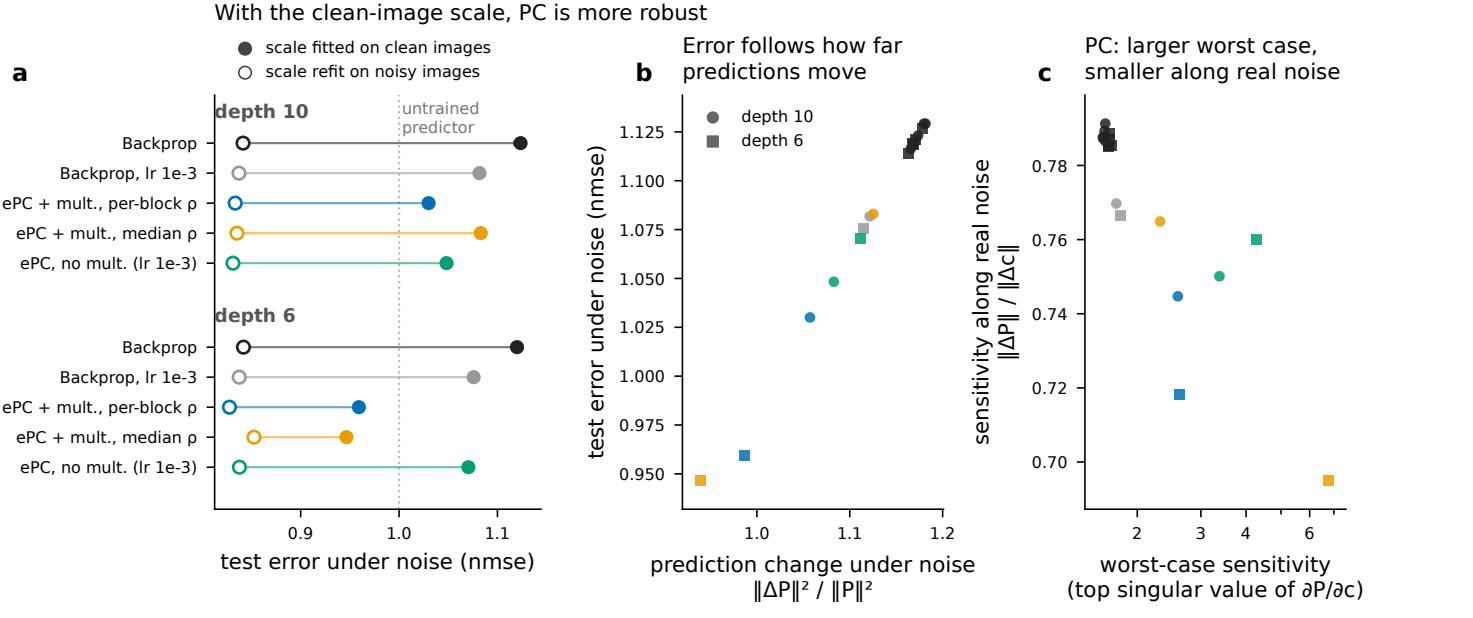
With the clean-image scale, PC is more robust
a
scale fitted on clean images
scale refit on noisy images
depth 10
untrained predictor
Backprop
Backprop, lr 1e-3
ePC + mult., per-block ρ
ePC + mult., median ρ
ePC, no mult. (lr 1e-3)
depth 6
Backprop
Backprop, lr 1e-3
ePC + mult., per-block ρ
ePC + mult., median ρ
ePC, no mult. (lr 1e-3)
0.9
1.0
1.1
test error under noise (nmse)
b
Error follows how far predictions move
depth 10
depth 6
test error under noise (nmse)
1.125
1.100
1.075
1.050
1.025
1.000
0.975
0.950
1.0
1.1
1.2
prediction change under noise
‖ΔP‖² / ‖P‖²
c
PC: larger worst case, smaller along real noise
sensitivity along real noise
‖ΔP‖ / ‖Δc‖
0.78
0.76
0.74
0.72
0.70
2
3
4
6
worst-case sensitivity
(top singular value of ∂P/∂c)
Backprop
Backprop, lr 1e-3
ePC + mult., per-block ρ
ePC + mult., median ρ
ePC, no mult. (lr 1e-3)